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Paper Code : PEC- CS801E/PECIT801D Internet of Things

UPID : 008270

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) Sound waves are similar to light waves in all aspects. True or False.
- (II) Mass sensors are a combination of SAW. True or False.
- (III) Semiconductor used in sensors will be _____
- (IV) Pyro electric coefficient of PVDF will be _____
- (V) Input signal to smart sensor is fed from _____
- (VI) Tungsten filament lamp is a _____ source of IR radiation.
- (VII) Quartz can be used as bio sensors. True or False.
- (VIII) Heater power in thick film hybrid technology is up to _____ mW.
- (IX) Permalloy are most suitable for _____
- (X) Non contacting type bio sensors are _____
- (XI) Polymer thick film can act as a sensing element. True or False.
- (XII) Basically, sound waves are _____

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What are the environmental monitoring types? [5]
3. What is an IoT Sensor? [5]
4. What are different types of smart sensors? [5]
5. Describe the advantages and disadvantages of smart sensors. [5]
6. Describe the layers of IoT protocol stack. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Explain the working principles of sensors. Describe different selection criteria for sensors. Give example. [5+7+3]
8. What are the physical properties, those a thermal sensor based on? Describe its applications categorically. [10 + 5]
9. Explain the Architecture of Smart Sensor. Identify the role of machine learning, artificial intelligence and data analytics in Internet of Things. Explain with example of electrode fabrication. [6+6+3]
10. Explain the interface electronic circuit for smart sensors. Why smart sensors are gaining importance? Identify the future scope of research in smart sensor. [6+4+5]
11. What is IoT-Based Environmental Monitoring? What are the essential components for the IoT based environmental monitoring? Describe the IoT architecture for environmental monitoring system. What is rain gauge in IoT? [2 + 5 + 6 + 2]

*** END OF PAPER ***